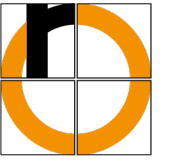


Computer Vision

Image Formation

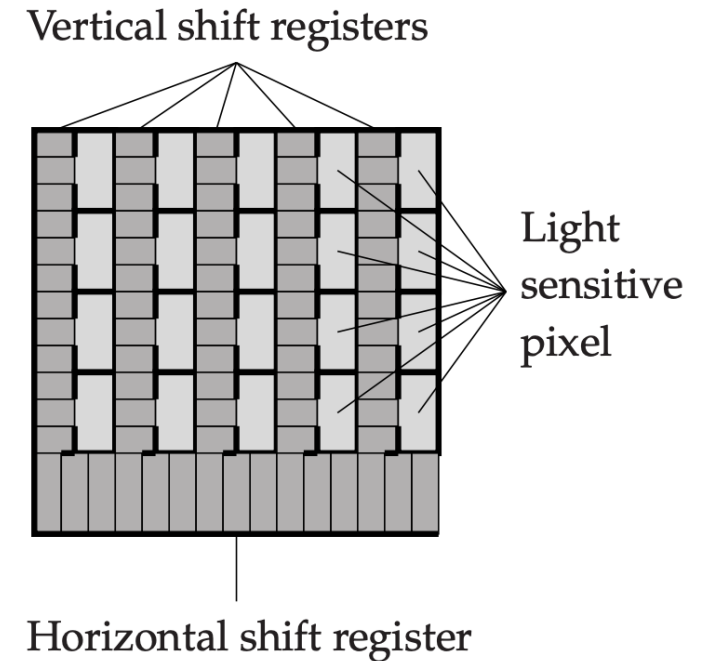
Technische Hochschule Rosenheim
Winter 2026/27
Prof. Dr. Jochen Schmidt



Sensors

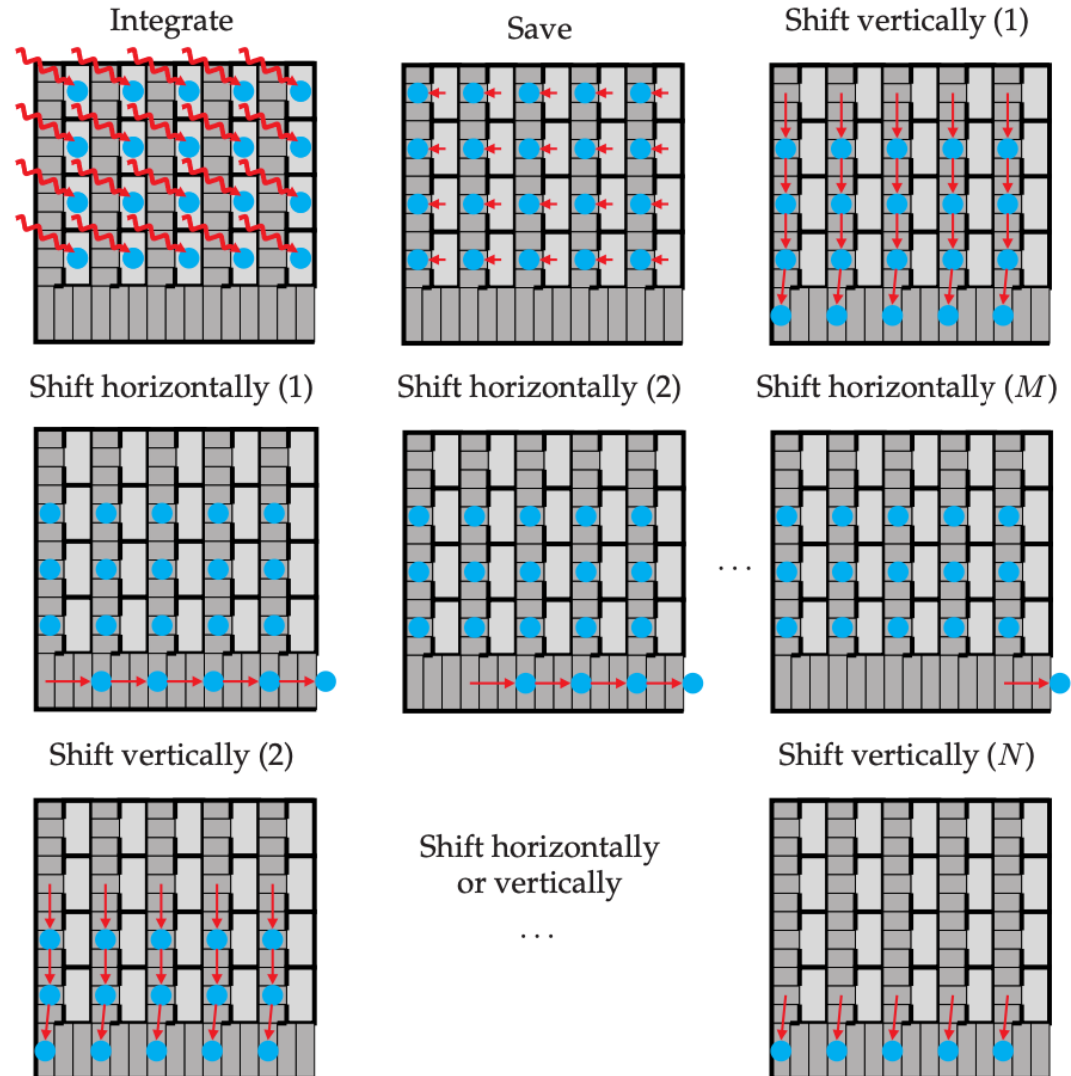
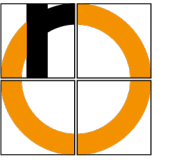
- Area sensor
 - e.g. CCD, CMOS chip
- Line sensor
 - Like area sensor, but pixels arranged in a line
- Point sensor
 - only records a single intensity value
- not only in the visible spectrum
 - also IR, UV, X-ray, radio waves (radar sensor), ...

- CCD = Charge Coupled Device
- consists of (interline sensor)
 - light-sensitive pixels, separated by bars and potential barriers
 - shift registers
 - not sensitive to light
 - take up about half of the chip area
- Full-frame sensor
 - uses almost the entire surface as a sensor
 - Charges are pushed through the light-sensitive area during readout
 - therefore mechanical shutter necessary (avoids exposure during readout)



from: [Beyerer16]

Interline CCD sensor



from: [Beyerer16]

Blooming



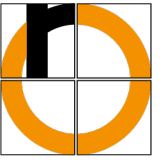
- is created during the exposure
- typically limited
- completely overexposed

Smear

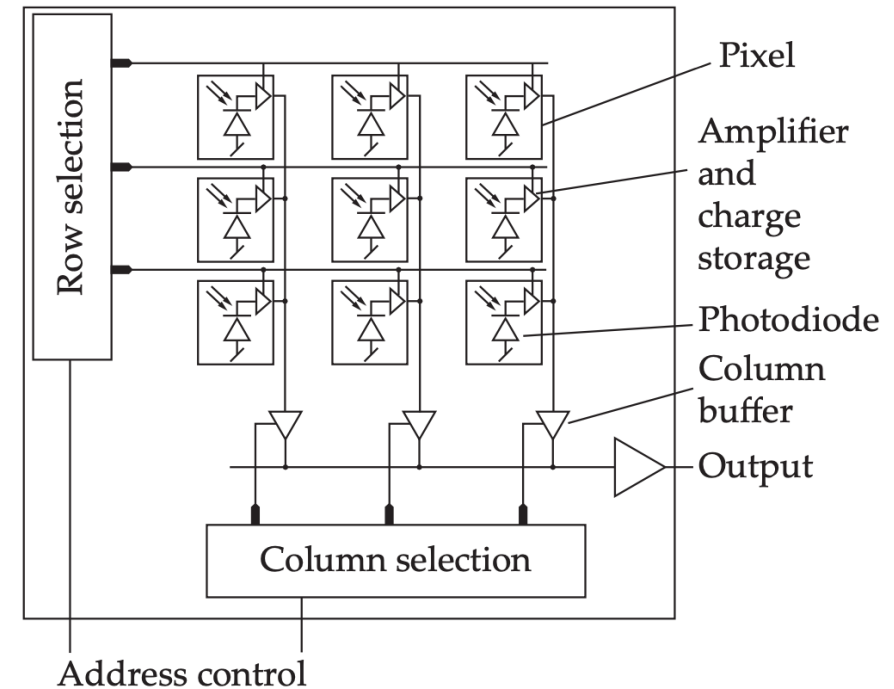


- is created during the readout
- Strips always up to the edge
- not completely overexposed

- inexpensive
- Low weight/size
- Insensitive to shocks and electromagnetic fields



- CMOS = Complementary Metal Oxide Semiconductor
- Detection of photons as with CCD
- Reading out the values
 - Direct control of individual pixels via transistors/address lines
 - Arbitrary image regions selectable
 - No shift registers necessary
- Advantages over CCD:
 - Control/processing electronics can be integrated directly on the chip
 - Free and random pixel access
 - Greater dynamic range how bright/dark the image can get
 - Low power consumption
 - Wide temperature range
 - No blooming, smear, shutter lag
- Disadvantages
 - less sensitive
 - higher noise



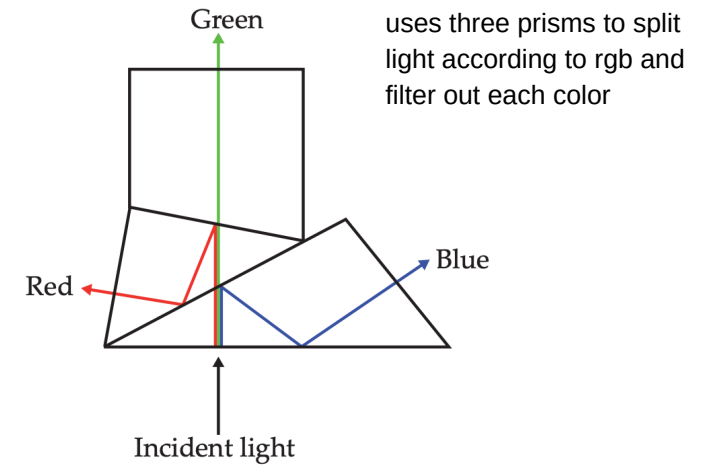
from: [Beyerer16]

- Light sensors in cameras ("imagers") detect the intensity of the light, (almost) regardless of the wavelength
- only gray images can be acquired

- Color information:

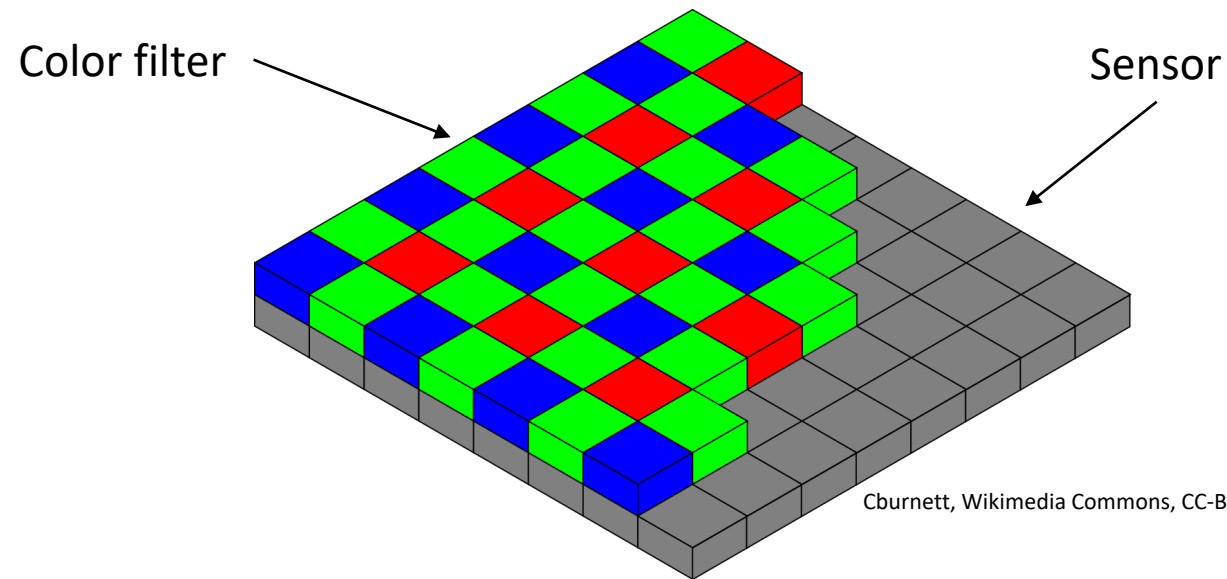
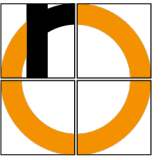
- either optical separation of the light according to wavelength, use 3 sensors each for R, G, B

Disadvantage: complex each color needs to hit the exact same coord
or leads to same effect as chromatic aberration via the lense



- or use of a color filter array (CFA), i.e. filters on each individual pixel
- Disadvantage: Loss of sensitivity due to filter**

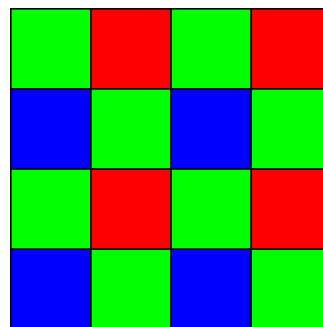
from: [Beyerer16]



Nyquist–Shannon sampling theorem
a continuous-time analog signal can be perfectly reconstructed from its discrete samples if the sampling rate is strictly greater than twice the highest frequency component of the signal.

Cburnett, Wikimedia Commons, CC-BY-SA-3.0

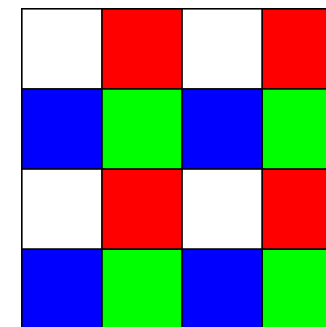
data format
png, tiff > lossless compression
dont use jpeg



Bayer (RGGB)

Cburnett, Wikimedia Commons, CC-BY-SA-3.0

green filters absorb less light,
so more green filters means
keeping more detail



RGBW

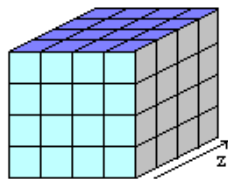
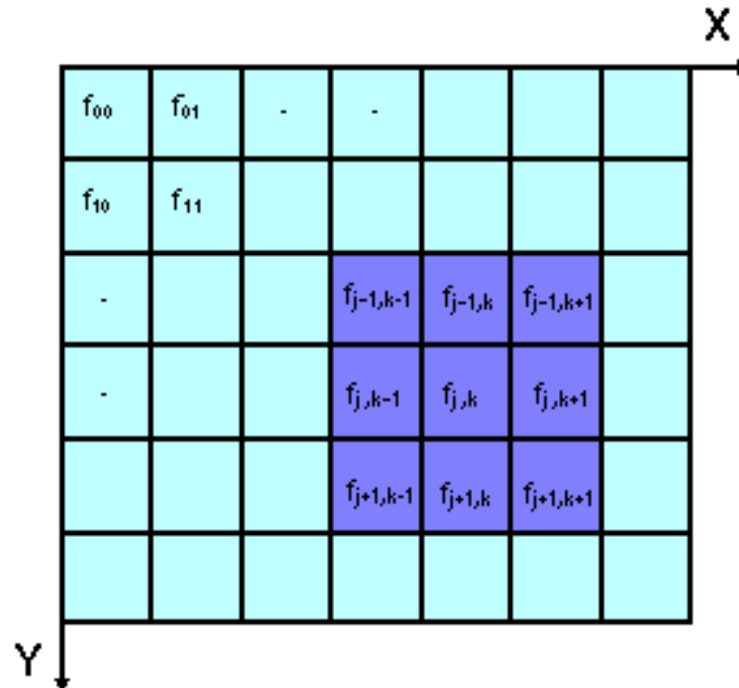
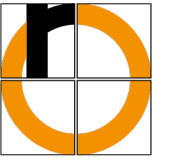
Dicklyon, Wikimedia Commons, Public Domain

no filter
no sensitivity loss
> combination of greyscale
and color channel
for specialized purpose

... of commercial CCD/CMOS cameras for machine vision

Number of pixels	640 x 480 up to 4800 x 3200
Pixel size	1.4 x 1.4 μm^2 up to 10 x 10 μm^2
Sensor size	¼" to 1", more rarely up to 35mm (mostly 4:3, diagonal from 4.5mm to 43.3mm)
Exposure time (<i>Belichtungszeit</i>)	10 μs to several seconds
Frame rate	3Hz to 200Hz
Color depth	8 bit to 16 bit per channel (internally often more)
Shutter (<i>Verschluss</i>)	global or rolling
Scanning Method	Interlace or Progressive Scan

linearly scans one row after another
interlace skips every other row, second pass scans remaining rows



Volume image

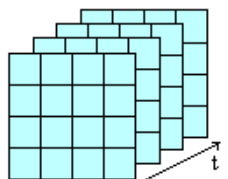
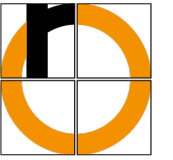
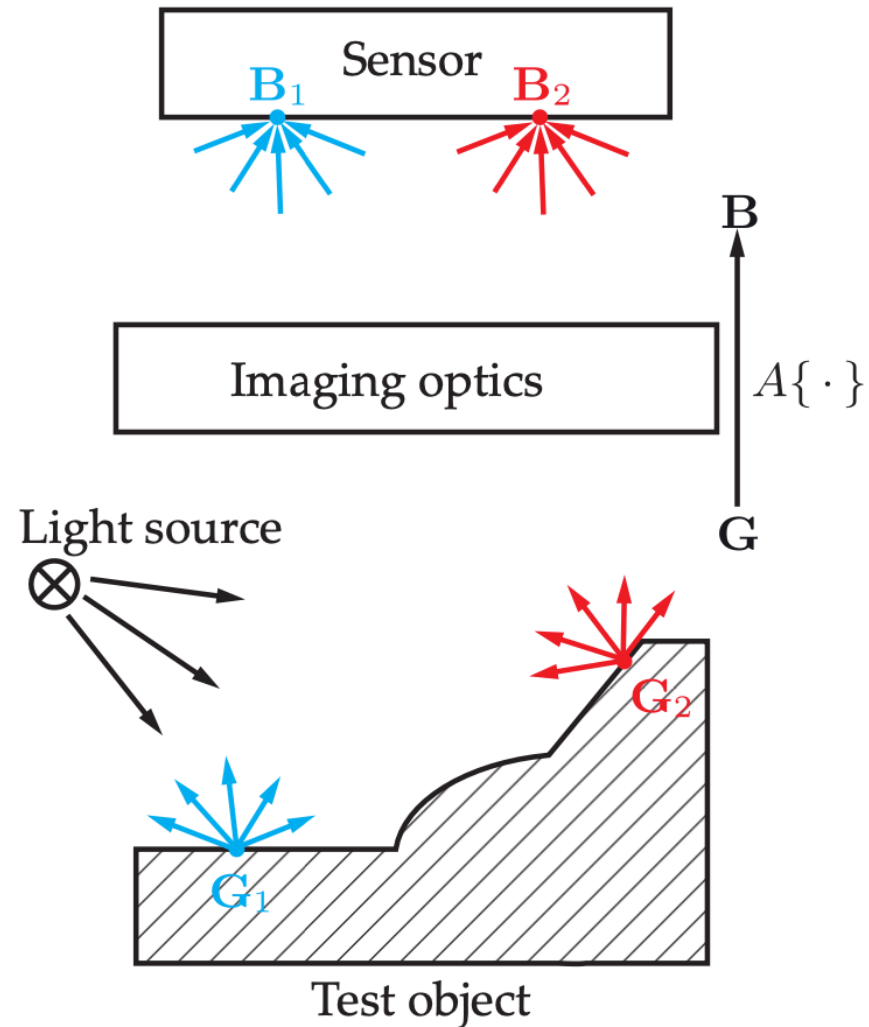
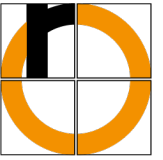


Image sequence

- Binary image
1 bit / pixel
- Gray value image
8 bit / pixel
10, 12, 16 bit / pixel
- Color image (RGB)
8 bit / color channel
10, 12, 16 bit / color channel
(often more internally)

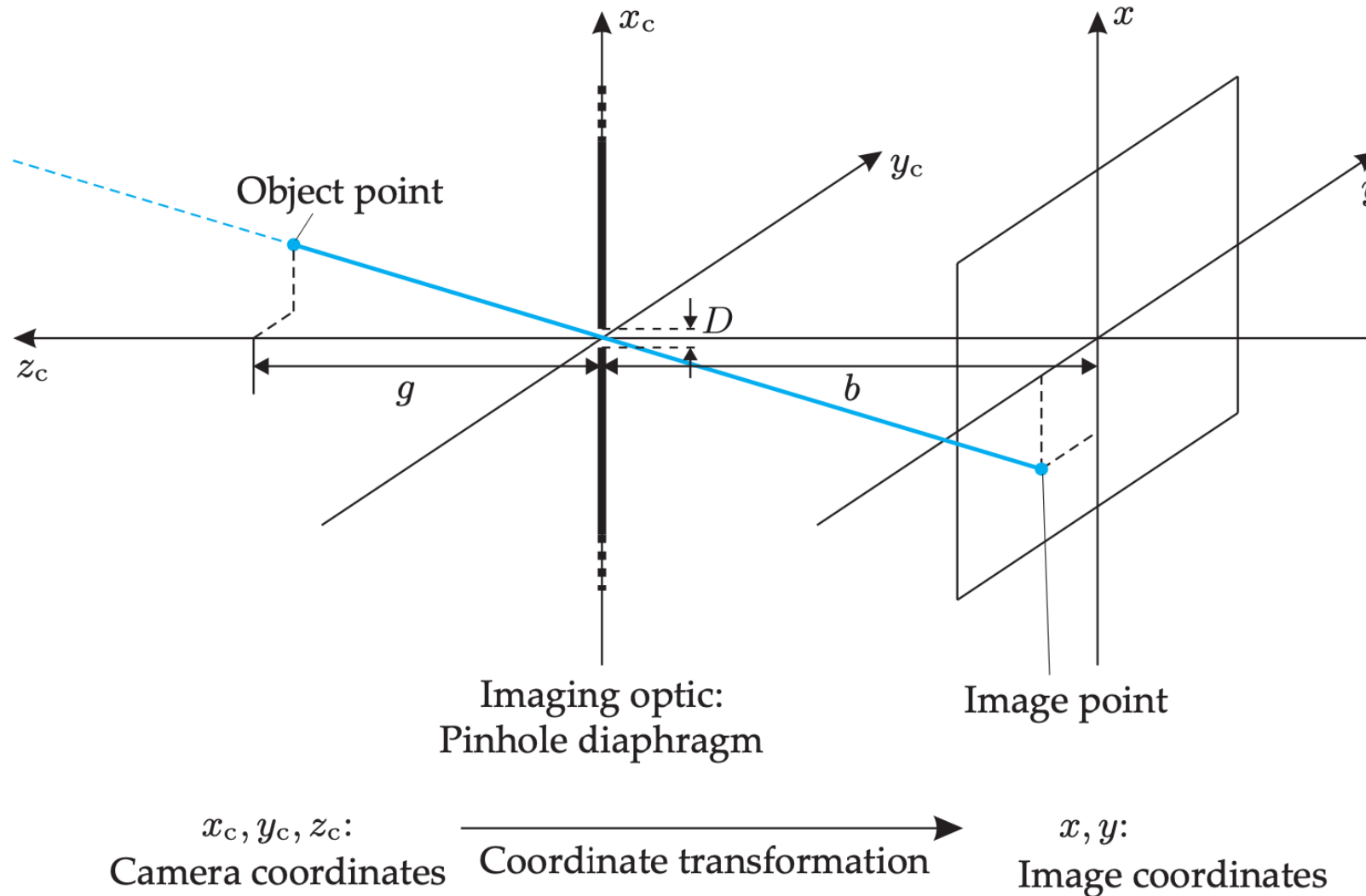
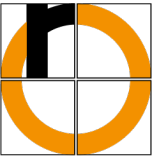


Optical Imaging



from: [Beyerer16]

Pinhole Camera Model (*Lochkameramodell*)

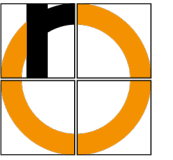


$$x = -\frac{b}{z_c} x_c$$

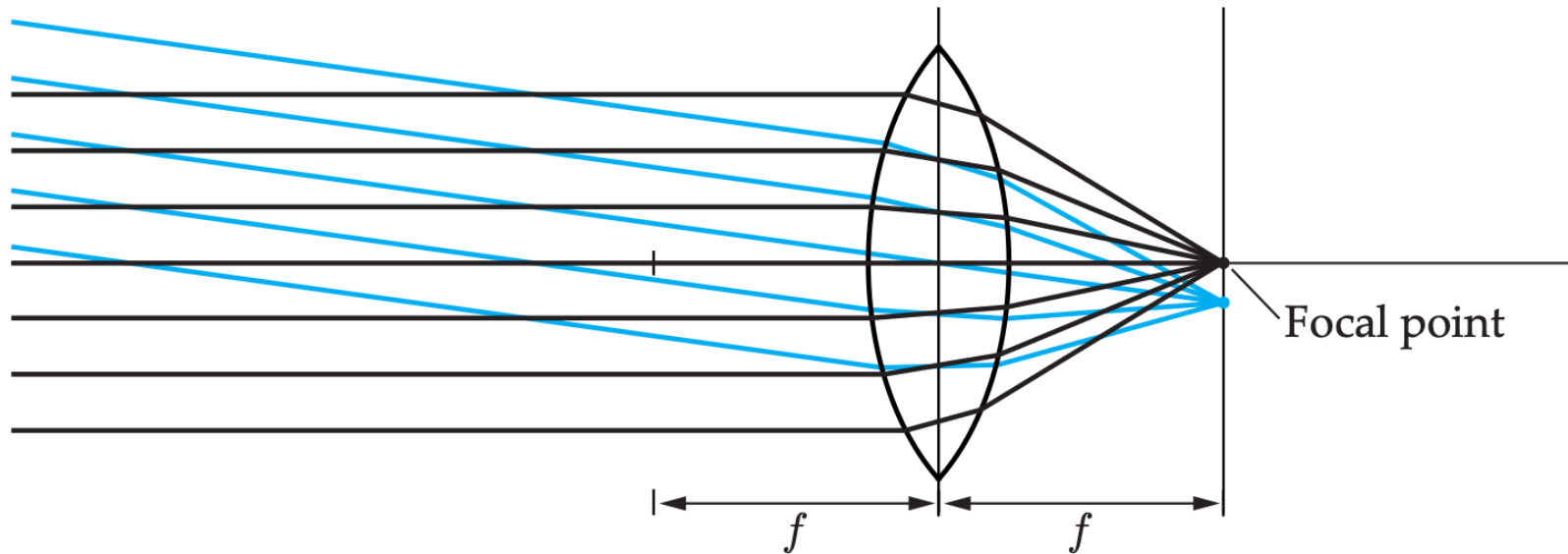
$$y = -\frac{b}{z_c} y_c$$

from: [Beyerer16]

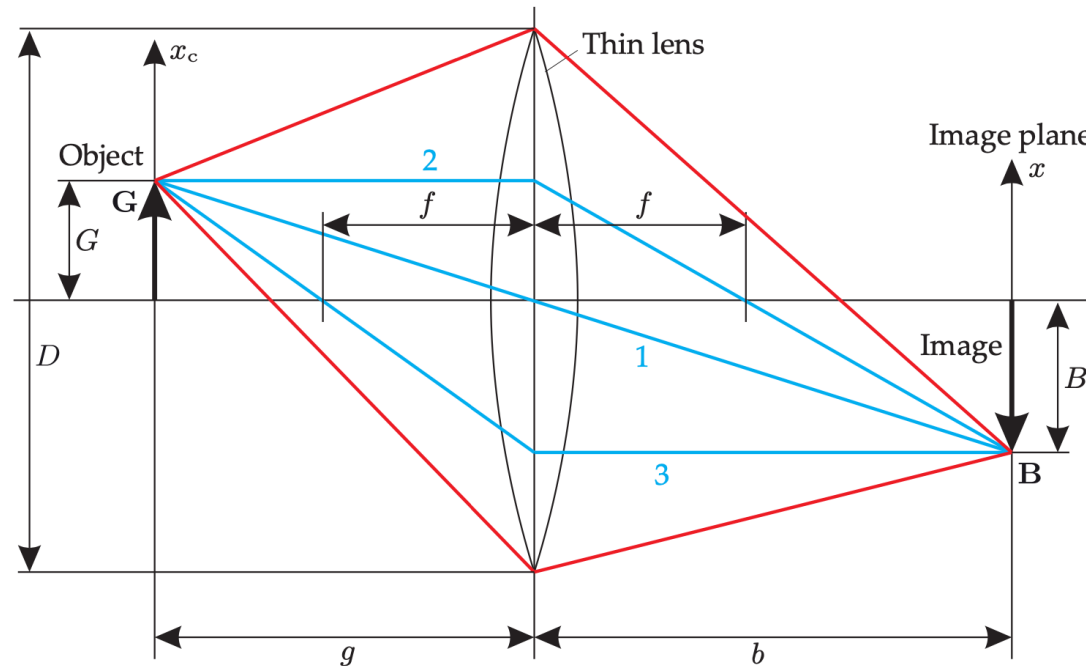
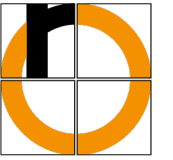
Imaging with a Single Lens



Parallel incident light beams are bundled in the focal plane



from: [Beyerer16]



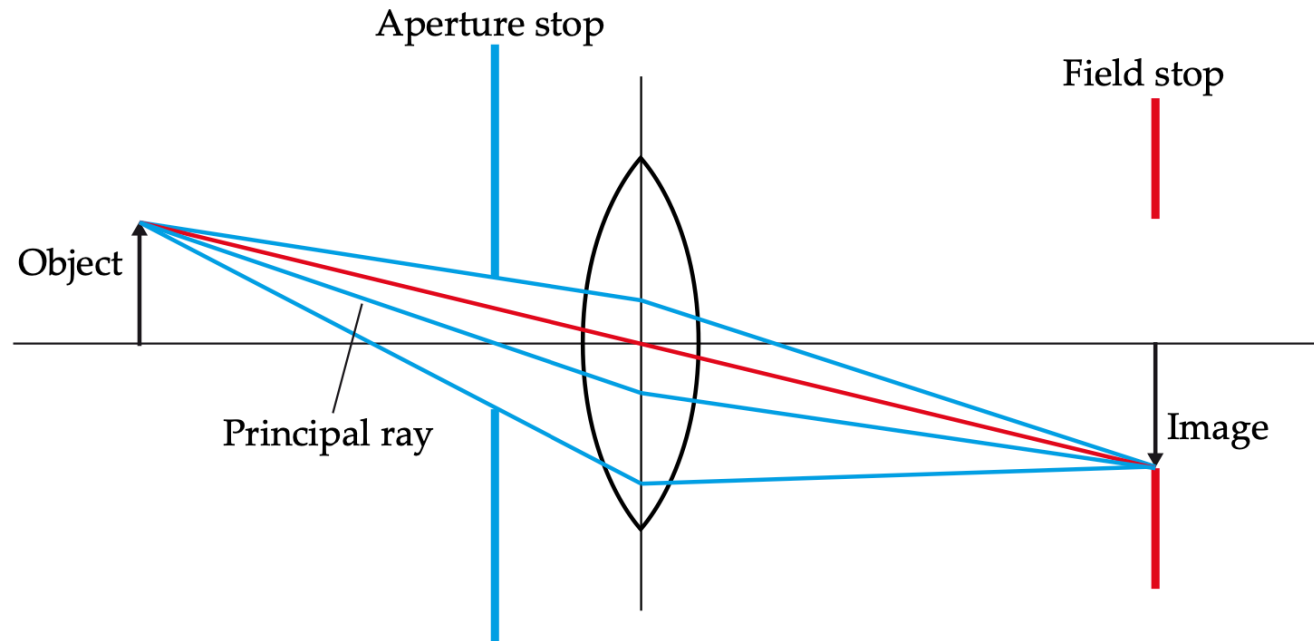
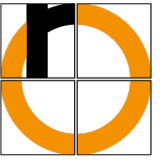
Imaging equation: $\frac{1}{f} = \frac{1}{g} + \frac{1}{b}$

Depth of field (*Schärfentiefe*) is limited

Only the plane at distance $g = \frac{bf}{b-f}$ is in focus on the image plane

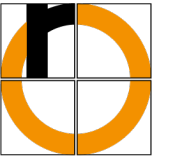
from: [Beyerer16]

Bundle Limitation: Aperture and Field Stop

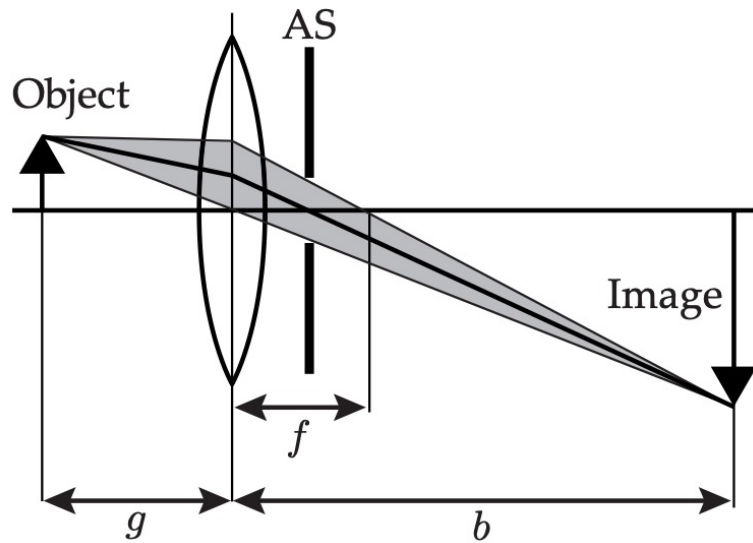


Field stop: (<i>Feldblende</i>)	limits the field of view/image field (determines image region)
Aperture stop: (<i>Aperturblende</i>)	limits the opening of the light beam position in front of/behind/between lenses determines the depth of field

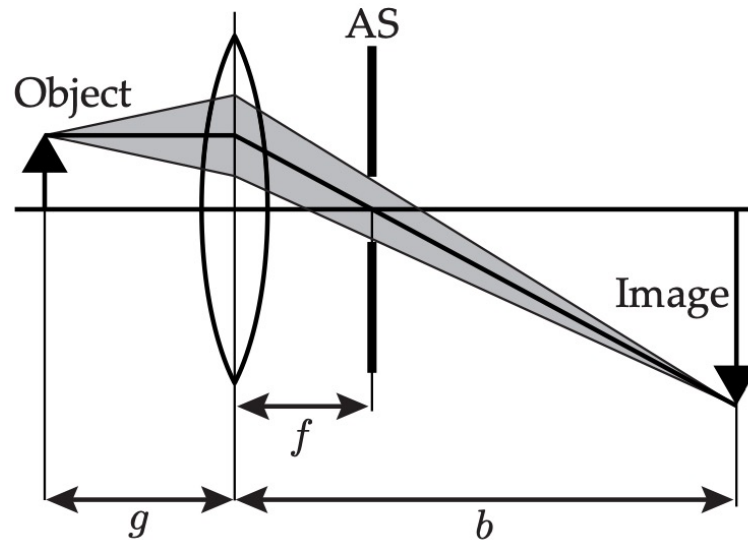
from: [Beyerer16]



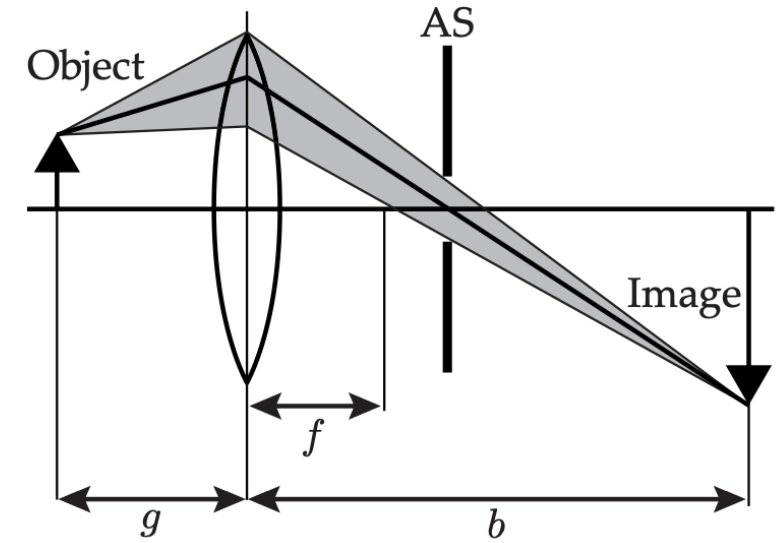
entocentric



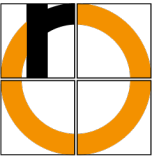
telecentric



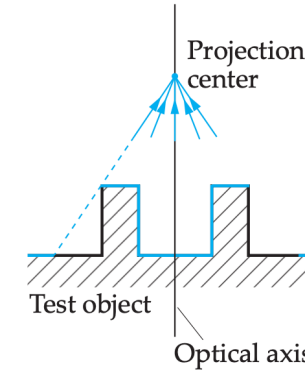
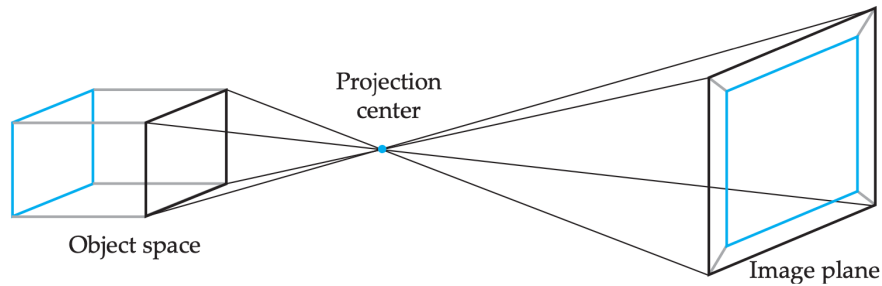
hypercentric



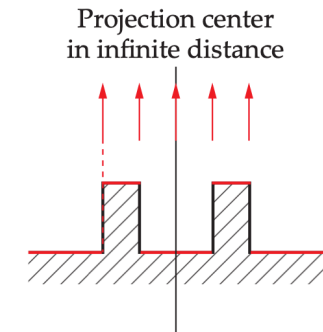
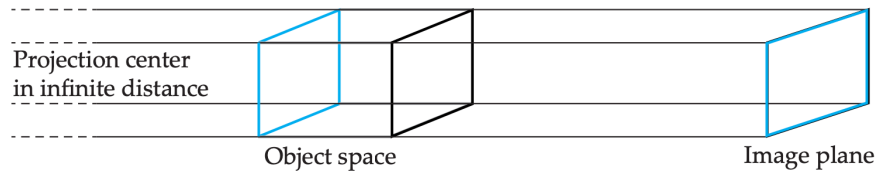
from: [Beyerer16]



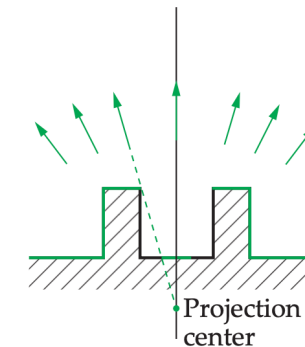
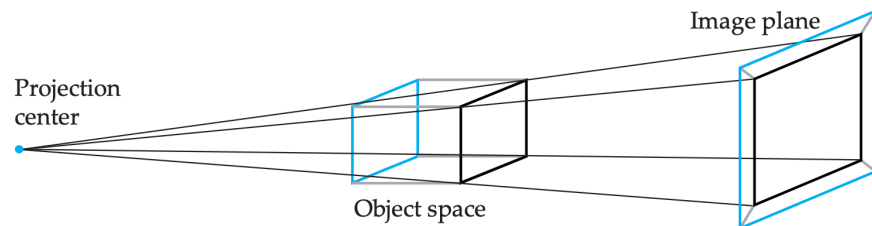
entocentric - the "normal" perspective projection/pinhole camera model



telecentric - also "parallel projection", "orthogonal projection"

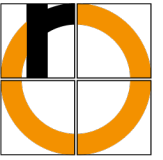


hypercentric



from: [Beyerer16]

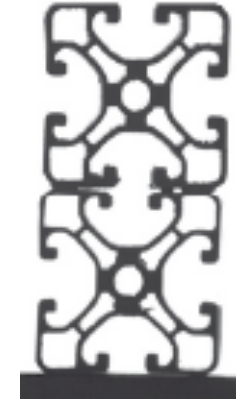
Example



Scene
(aluminum profiles)



entocentric

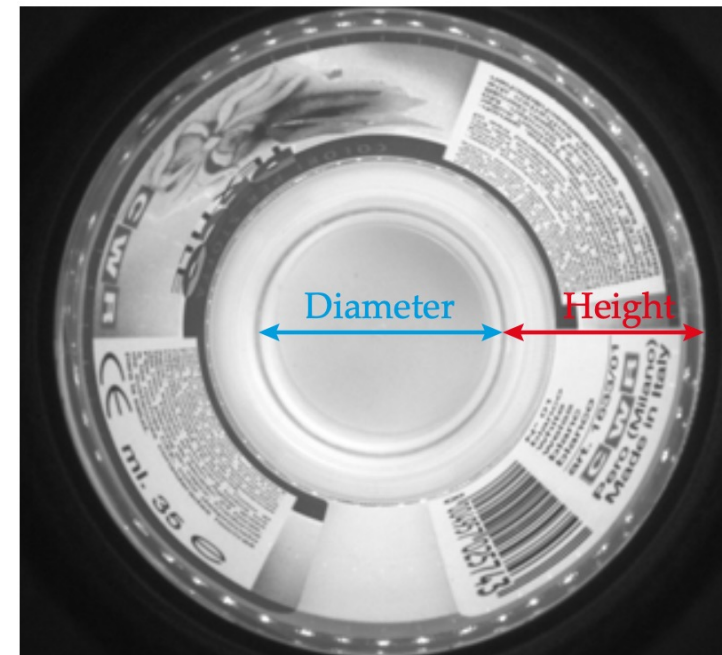
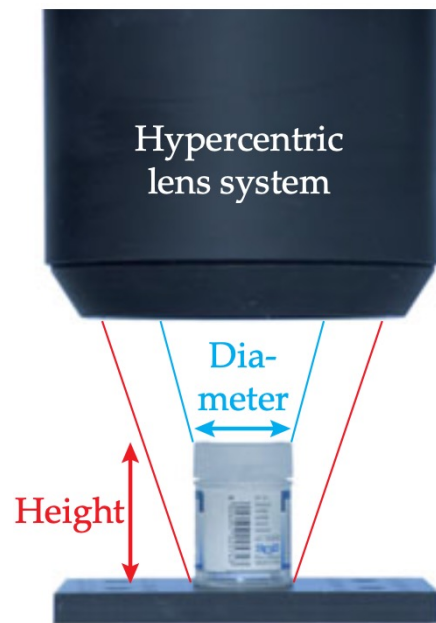


telecentric

telecentric/hypercentric: lens must be larger than the object
Telecentric lenses are available up to approx. 30cm diameter

from: [Beyerer16]

Example of a Hypercentric Perspective



from: [Beyerer16]

- Perspective
- Focal length (*Brennweite*)
 - fixed (which?) or zoom (which range? manual/motor?)
 - suitable for the image sensor
- Size matching the image sensor
- Working distance (Arbeitsabstand)
- Lens speed (*Lichtstärke*)
 - depending on aperture stop and quality of optics
- Resolution
 - suitable for the sensor
 - measured in line pairs/mm (lp/mm)
- Lens mount (*Objektivanschluss*) suitable for the camera
- Filter thread necessary? (*Filtergewinde*)
- Quality of the optics – imaging errors
 - geometric (e.g. radial distortions)
 - chromatic (refractive index depends on wavelength)

[Beyerer16] Beyerer, J., Puente Leon, F., Frese, Ch.: ***Machine Vision***, Springer, 2016.